



AMERICAN INTERNATIONAL UNIVERSITY–BANGLADESH (AIUB)

Department of Computer Science
Faculty of Science and Technology

CSC2210: Object Oriented Programming 2

Project Report On
PET SHOP MANAGEMENT SYSTEM

Developed Using C# Windows Forms and SQL Server Database

Submitted By:
[Group Members Information]

Supervised By:
[Instructor Name]

Table of Contents

Chapter 1: Introduction

Chapter 2: Project Overview and Objectives

Chapter 3: User Stories and Functional Requirements

Chapter 4: System Design and Database Design

Chapter 5: Implementation Details

Chapter 6: Graphical User Interface Description

Chapter 7: Testing and Validation

Chapter 8: Conclusion and Future Improvements

References

Chapter 1: Introduction

The Pet Shop Management System is a desktop-based application developed to automate the daily activities of a pet shop.

The system provides separate facilities for administrators and customers. It manages pet products, inventory, customer accounts, orders, payments, and customer feedback. The application is developed using C# Windows Forms with SQL Server database integration.

The main purpose of this project is to reduce manual management effort and provide an organized platform for maintaining pet shop operations.

The project follows object-oriented programming concepts including class-based design, form-based interaction, database connectivity, and modular development.

Chapter 2: Project Overview and Objectives

Project Name: Pet Shop Management System

Objectives:

- Manage pet products, food items, accessories, medicine, and related inventory.
- Provide secure login and account creation facilities.
- Allow customers to browse products and purchase items.
- Generate payment records and cash memos.
- Maintain customer reviews and feedback.
- Provide administrators with inventory and sales management features.

Chapter 3: User Stories and Functional Requirements

Administrator

The administrator controls the overall system. Admin features include:

- Admin login authentication.
- Inventory management.
- Adding, updating, and removing products.
- Monitoring sales information.
- Managing customer reviews and system data.

Customer

Customer features include:

- Creating an account and logging into the system.
- Browsing available products.
- Adding products for purchase.
- Completing payment procedures.
- Receiving cash memo/order information.
- Submitting reviews.

Chapter 4: System Design and Database Design

The application uses Microsoft SQL Server as the database management system.
The C# application connects with the database through SQL Server connection strings.

Main database entities include:

- Users/Customers
- Products
- Categories
- Orders
- Payments
- Reviews
- Inventory records

The database is designed to store product information, user credentials, transactions and feedback efficiently.

Chapter 5: Implementation Details

The project contains multiple C# Windows Forms modules:

LoginForm:

Provides authentication interface for system users.

CreateAccount:

Allows new customers to register with personal information.

AdminDashboard:

Provides administrative controls and monitoring facilities.

AdminInventoryManagement:

Handles product stock, categories, prices and inventory updates.

CustomersDashboard:

Displays available products, shopping cart operations and customer interaction.

CustomerPaymentCashMemo:

Generates payment details and purchase records.

GiveReview:

Allows customers to submit reviews.

Chapter 6: Graphical User Interface Description

The application provides a user-friendly graphical interface.

Different dashboards are created for administrators and customers.

The interface contains navigation panels, product displays, forms, buttons, images and data presentation components.

The project resources include pet images, product images, wallpapers and logos to improve usability and visual appearance.

Chapter 7: Testing and Validation

Testing was performed to verify the major functionalities of the system.

Login Testing:

Verified correct authentication and invalid credential handling.

Inventory Testing:

Verified product addition, modification and stock updates.

Payment Testing:

Verified order processing and cash memo generation.

Database Testing:

Verified insertion, retrieval and updating of records between application and SQL Server.

Chapter 8: Conclusion and Future Improvements

The Pet Shop Management System successfully demonstrates a real-life desktop application using C# and database integration.

The system improves operational efficiency by organizing product management, customer purchasing and transaction records.

Future improvements may include:

- Online payment gateway integration.
- Cloud database support.
- Mobile application version.
- Advanced sales analytics and reporting.
- Customer notification system.

ScreenShots:



References

1. Microsoft C# Documentation.
2. SQL Server Database Documentation.
3. Object Oriented Programming course materials.